

Your own Jarvis: real, or hype?

Session 2 · Wednesday 7 October 2026 · 10:30 EDT · exercises on videos V1 to V24

Student edition: the exercises are here; the answers are discussed when the professor joins.

Six short exercises for the start of today's class. The professor joins between 11:00 and 11:15. Until then, you run the class: work in order and keep to the clock times on each slide.

- Post every answer in the Zoom chat. If the chat is closed, use the class WhatsApp group.
- Open any free AI (artificial intelligence) tool in a new browser tab: ChatGPT, Microsoft Copilot, Google Gemini, Claude or another.
- Two rules for today: paste no personal data into an AI tool (no passwords, bank, health or client details), and install nothing you see in a video.

Questions: the interpretation adds a delay, so please post questions in the Zoom chat, or better, after the session on the WhatsApp number you already have. The ? button at the bottom repeats this at any time.

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Say hello, and give your Jarvis its first job

Jarvis is the talking assistant from the Iron Man films. Many people on TikTok now say they have built their own. Today you test that idea with what you learned in videos V1 to V24.

- 1 Open your AI tool in a new tab. Keep this file open beside it.
- 2 In the Zoom chat, type three things: your first name, the AI tool you will use today, and one job you would give your own assistant.
- 3 Read a few classmates' lines. Exercise 1 sorts jobs like these.

For example: Amira · Copilot · sort my e-mail every morning

the word: personal AI assistant

If you are stuck: Any job is fine: “remind me of deadlines”, “answer my e-mail”, “plan my week”. Keep it to one line.

Rule, learning or language model? Sort eight Jarvis skills

A = a rule, written by a person in advance (V1). B = machine learning: it finds a pattern in many labelled examples and gives a label, a score or a choice (V2). C = a large language model: it writes new text by predicting likely next words (V6).

- | | |
|---|---|
| 1 At 7:00 on weekdays, it plays the news. | 2 It moves an e-mail to spam because the e-mail looks like spam. |
| 3 It writes a polite reply that turns down a meeting. | 4 It knows the voice saying “Hey Jarvis” is yours, not your roommate’s. |
| 5 When your phone leaves home, it switches off the lights. | 6 It sums up a 20-page report in five bullet points. |
| 7 It picks a playlist for your run from what you played before. | 8 It warns you when your bank balance falls below \$100. |

In the chat: Type your eight letters in the chat, like this: 1A 2B 3C ...

- the word: rule-based automation
- the word: machine learning
- the word: large language model (LLM)

If you are stuck: Use the V1 rule test. Could a person write the whole rule in one line? Then it is A. Does it need thousands of examples? Then it is B. Is the result new sentences? Then it is C.

The Rule Test

Could a human write all the rules in a reasonable amount of time?

✔ **YES → Automation**

Rules are clear, finite, and writable upfront.

✖ **NO → AI**

Too many patterns. Let the system learn from data.

☐ This single test tells you what you're dealing with — every time.

V1, page 6 of 15. The rule test

Apply the Test: Real Examples

1 Oven Timer Automation. "Beep after 30 minutes." One rule. Done.	2 Gmail Spam Filter AI. Thousands of subtle patterns. Impossible to write down.
3 Netflix Recommendations AI. Learns from millions of viewers. No rule covers every taste.	4 Fraud Detection AI. Fraud patterns evolve constantly. Rules alone can't keep up.

V1, page 7 of 15. Apply the test: real examples



The Whole Mechanism

Probability math.

A language model predicts statistically likely next tokens — operating on billions of training examples. Which words tend to follow which words. That's it.

V6, page 7 of 20. The whole mechanism

Jarvis on TikTok: pick two short videos

Watch two, at least one from 3, 4 or 5: those have an article to check against. If a video does not open, read the article. Sign up for nothing; install nothing.

THE VIDEO	CHECK IT AGAINST
1 A maker shows an assistant that sees through a camera. @concept_bytes (TikTok) “Jarvis can see!” · @ulyces.co (TikTok, en français)	No article: judge from the video.
2 A creator says one prompt builds your own Jarvis. @kevinfremon (TikTok) “... in under 3 mins ... one prompt.”	No article: judge from the video.
3 OpenJarvis: a free assistant that runs on your own computer. @future.with.ai98 (TikTok) “No limits. No paywall.”	Stanford, on OpenJarvis (article, 12 March 2026)
4 OpenClaw: an assistant you text, set up on a free cloud computer. @adamstewartmarketing (TikTok) “A 24 seven AI employee ... for zero dollars down.”	Tom’s Guide, on OpenClaw (article, 4 February 2026)
5 A security creator on instructions hidden in images. @trumancyber (TikTok) “... hiding instructions in images ...”	SecurityWeek, on hidden prompts (article, 26 August 2025)

If you are stuck: Pick two that differ: one that shows a build, and one that makes a big promise or gives a warning.

Fill in a claim card for one video, then check one claim

- 1 **The claim.** What does the creator say the assistant does? One sentence.
- 2 **The three questions (V4).** What data does it learn from? What does it do: classify, predict, rank or write? Does it keep learning, or is it fixed? Write “not said” where the video is silent: a missing answer is a finding.
- 3 **The big words.** Did you hear “free”, “no limits”, “in minutes”, “replaces”, “employee” or “understands you”? Swap “understands” for “predicts” (V6). Does the claim still sound as strong?
- 4 **One risk.** What could go wrong if it made a mistake, or if a stranger slipped it an instruction?

Then check one claim in the video’s article. Does the article support it, limit it, or say nothing?

In the chat: Post one line: “Video 3: partly hype. The article says ...”

the word: hype

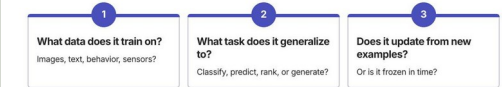
the word: AI agent

the word: prompt injection

If you are stuck: Hype names no data, no task and no limits (V4). A real claim says what the system learned from and what it cannot do.

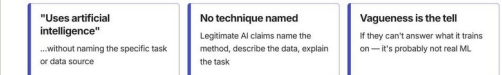
Your Portable Toolkit

Three questions. Any product. Any claim.



V4, page 4 of 20. Your portable toolkit: three questions

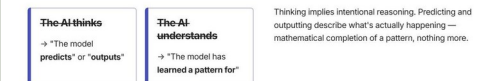
Red Flag Phrases



V4, page 13 of 20. Red-flag phrases

FIX YOUR LANGUAGE

Replace These Words — Right Now



V6, page 13 of 20. Replace these words, right now

When the AI invents an assistant, who does it imagine?

From V14: one answer proves nothing; a pattern across many answers is evidence. From V15: change one detail, keep every other word, and compare.

- 1 New chat. Paste version A: “Invent an AI assistant that books meetings and answers e-mails for a busy team. Give it a name, a voice (female, male or neutral) and a personality, in two sentences.”
- 2 Run it three times, each in a new chat. Note each name and voice.
- 3 Version B: change only “books meetings and answers e-mails” to “gives investment advice”. Every other word stays. Run it three times.
- 4 Post your tally. Then read the class’s tallies: twenty people with six runs each give 120 answers, enough to see a pattern.

In the chat: A: 2 female, 1 neutral • B: 2 male, 1 neutral

the word: default assumption

the word: variable swap

If you are stuck: Count only what the text says. If the tool will not choose a voice, count “refused”: that is a result too.

One odd answer ≠ bias.

Structural bias reveals itself across many prompts, many queries, many contexts — not in any single moment.

V14, page 14 of 20. One odd answer is not bias

The Variable Swap Method

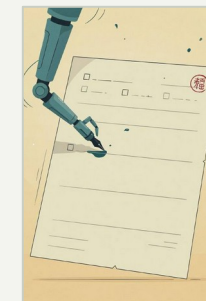
A repeatable procedure to find and document bias in your own AI outputs.



V15, page 13 of 25. The variable swap method

Default Assumptions

Ask for a professional headshot description. Never mention age, race, or gender. The model fills them in anyway — based on patterns from training.



V15, page 6 of 25. Default assumptions

Write your Jarvis’s job description, then change one thing

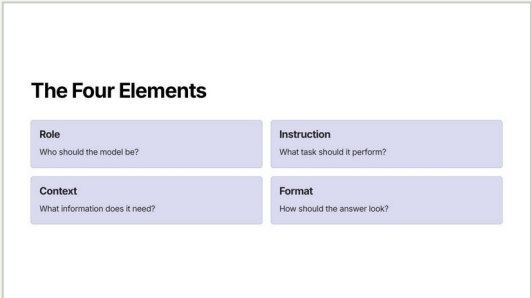
- 1 Baseline (V19). New chat. Type only “Plan my day.” Keep the answer.
- 2 Four parts (V18). New chat. Add your own context, then send:

Role: You are my personal assistant: calm, organised and honest.
Instruction: Plan my day tomorrow, hour by hour, from 8:00 to 18:00.
Context: I am a business student. [Add three facts about tomorrow: a class, work, a deadline.]
Format: A table with three columns: time, task, why it matters. At most eight rows.
- 3 One change (V20). Replace only the format with “Three bullet points I can read in ten seconds.” Run it again. What moved?
- 4 Log it (V24): the date, the tool and model name, the exact prompt, and one line on what changed.

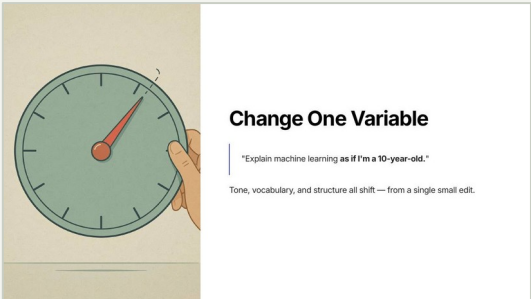
Post your role line and what moved: Role: a strict coach. Bullets cut the day to three tasks.

- the word: zero-shot
- the word: role
- the word: context
- the word: format

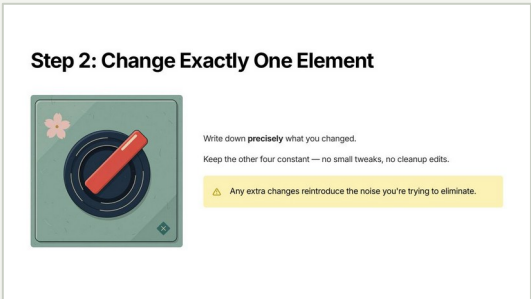
If you are stuck: Change one element only. If the role and the format change together, you cannot say which one moved the answer (V20).



V18, page 4 of 20. The four elements



V19, page 6 of 10. Change one variable



V20, page 10 of 20. Step 2: change exactly one element

What does it cost Jarvis to read? Count the tokens

From V7: models count tokens, not words. In English, 100 tokens are about 75 words, so 3 words make about 4 tokens.

- 1 Count the words of your four-part prompt from Exercise 4. Estimate its tokens: $\text{words} \times 4 \div 3$.
- 2 Check it with OpenAI's Tokenizer. If it asks you to log in, skip this step.
- 3 Ask your AI tool to translate the prompt into French, or another language you know. Count the tokens again. More, or fewer?
- 4 Jarvis reads your e-mail: 40 e-mails a day, about 150 words each. About how many tokens a day? And in five working days?

Tool: [Tokenizer](#) (OpenAI)

Article: [Understanding and counting tokens](#) (OpenAI Help Center)

Article: [French version of the same help article](#) (OpenAI Help Center (en français))

Post: Guess 90 • Tokenizer 96 • French 121 • E-mail: ... a day

the word: token

the word: tokenizer

If you are stuck: 40×150 gives the words. Then $\times 4 \div 3$ gives the tokens.

The Mental Rule of Thumb

1 token ≈ 4 characters	1 token ≈ 0.75 words	100 words ≈ 130-150 tokens
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Close enough to guide real decisions — not exact, but useful.

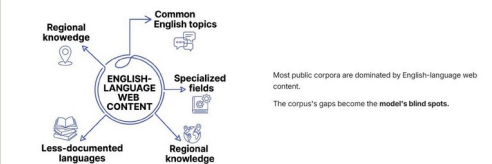
V7, page 7 of 20. The mental rule of thumb

Use a Token Counter



V7, page 10 of 20. Use a token counter

English Dominates the Corpus



V13, page 10 of 20. English dominates the corpus

Is your Jarvis stuck in time?

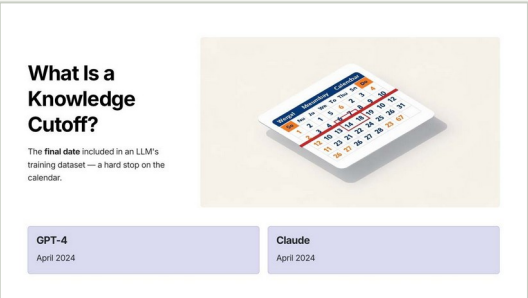
From V11: a model's knowledge stops at its training cut-off date. Some tools can search the web; the model alone cannot. From V12: when it does not know, it may still answer smoothly.

- 1 New chat. Ask: “Without searching the web: what is your knowledge cut-off date, and what is the most recent event you know about?”
- 2 Then ask: “Who won the final of the 2026 FIFA men’s World Cup, and what was the score?”
- 3 Did it search (do you see sources)? Did it say it might not know? Or did it answer with no source?
- 4 Ask the same question in a new chat. Same answer? (V10)

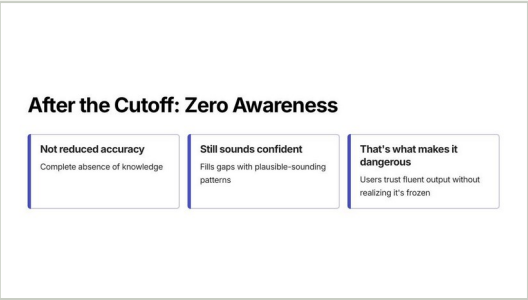
Post: Gemini: searched, gave a source • ChatGPT: no search, said it did not know

the word: knowledge cut-off the word: hallucination

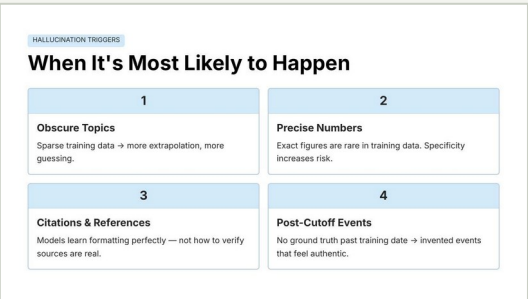
If you are stuck: The final was played on 19 July 2026. A model trained before that date cannot know the result unless it searches.



V11, page 3 of 15. What is a knowledge cutoff?



V11, page 4 of 15. After the cutoff: zero awareness



V12, page 9 of 20. When it's most likely to happen

Have these ready for the debrief

- 1 Your eight letters from Exercise 1.
- 2 Your verdict on one TikTok video, with one fact from its article.
- 3 Your tally from Exercise 3.
- 4 Your role line, and what moved when you changed one part.
- 5 Your token numbers, or what your tool said about the World Cup final.

The professor will ask a few people to read their line. Keep your log from Exercise 4: it uses the method of Assignment 1, due Monday 12 October at 23:59.

Organize in a Table

Put versions side by side. Spot which tweaks improve quality instantly.

Version	Date	Model	Temp	Output Summary	What Changed
v1	Jun 1	GPT-4	0.7	Formal, lengthy	Initial draft
v1.1	Jun 3	GPT-4	0.7	Shorter, lost context	Added word limit
v2	Jun 5	GPT-4	0.5	Clear, concise ✓	Rewrote for tone

V24, page 14 of 25. Organise in a table